1.1.62

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Question

If F(s) is the Laplace transform of function f(t), then Laplace transform of $\int_0^t f(\tau)d\tau$ is

1. $-F(s)$
2. $-F(s) - f(0)$
3. $\frac{1}{s}F(s)$
4. $\int F(s)ds$

Solution

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}dt \tag{1}$$

Let,

$$g(t) = \int_0^t f(\tau)d\tau \tag{2}$$

$$\mathcal{L}\{g'(t)\} = \int_0^\infty g'(t)e^{-st}dt \tag{3}$$

$$= g(t)e^{-st}\Big|_0^\infty - \int_0^\infty g(t)\frac{d}{dt}(e^{-st})dt \tag{4}$$

Solution

$$= (0 - g(0)) + s\int_0^\infty g(t)e^{-st}dt \tag{5}$$

$$g(0) = \int_0^0 f(\tau)d\tau = 0 \tag{6}$$

$$\implies \mathcal{L}\{g'(t)\} = \frac{1}{s}\mathcal{L}\{g(t)\} \tag{7}$$

$$\mathcal{L}\{g(t)\} = \frac{1}{s}\mathcal{L}\{g'(t)\} \tag{8}$$

Solution

Applying fundamental theorem of calculus to (2),

$$\frac{d}{dt}g(t) = f(t) \tag{9}$$

So we finally end up with

$$\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \frac{1}{s}\mathcal{L}\{f(t)\} = \frac{F(s)}{s} \tag{10}$$

$$\boxed{\textbf{Option C is the right answer}}$$

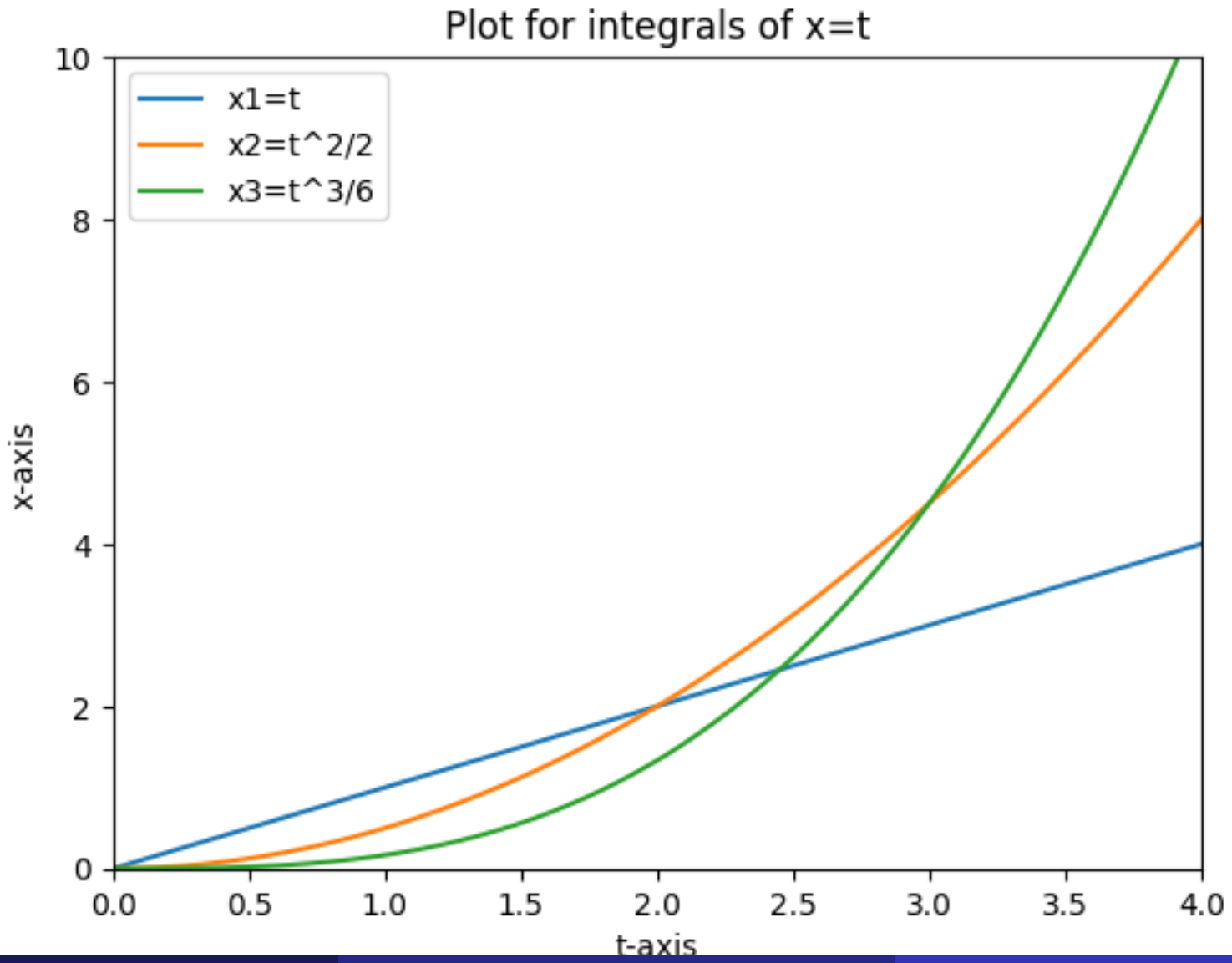
Python Code - Plotting time-domain signals (successively integrated) I

```
from matplotlib import legend
import numpy as np
import matplotlib.pyplot as plt
from scipy import integrate

x = np.linspace(0, 1000, num=100000)
y0 = np.ones(100000)

y1 = integrate.cumulative_simpson(y0, x=x, initial = 0)
y2 = integrate.cumulative_simpson(y1, x=x, initial = 0)
y3 = integrate.cumulative_simpson(y2, x=x, initial = 0)

plt.plot(x, y1, label = "x1=t")
plt.plot(x, y2, label = "x2=t^2/2")
plt.plot(x, y3, label = "x3=t^3/6")
plt.legend()
```

Python Code - Plotting time-domain signals (successively integrated) II

```
plt.xlabel("t-axis")
plt.xlim(0,4)
plt.ylabel("x-axis")
plt.ylim(0,10)
plt.title("Plot for integrals of x=t")
```

Plot for input time-domain signals $f(t) = t, f(t) = t^2/2, f(t) = t^3/6$



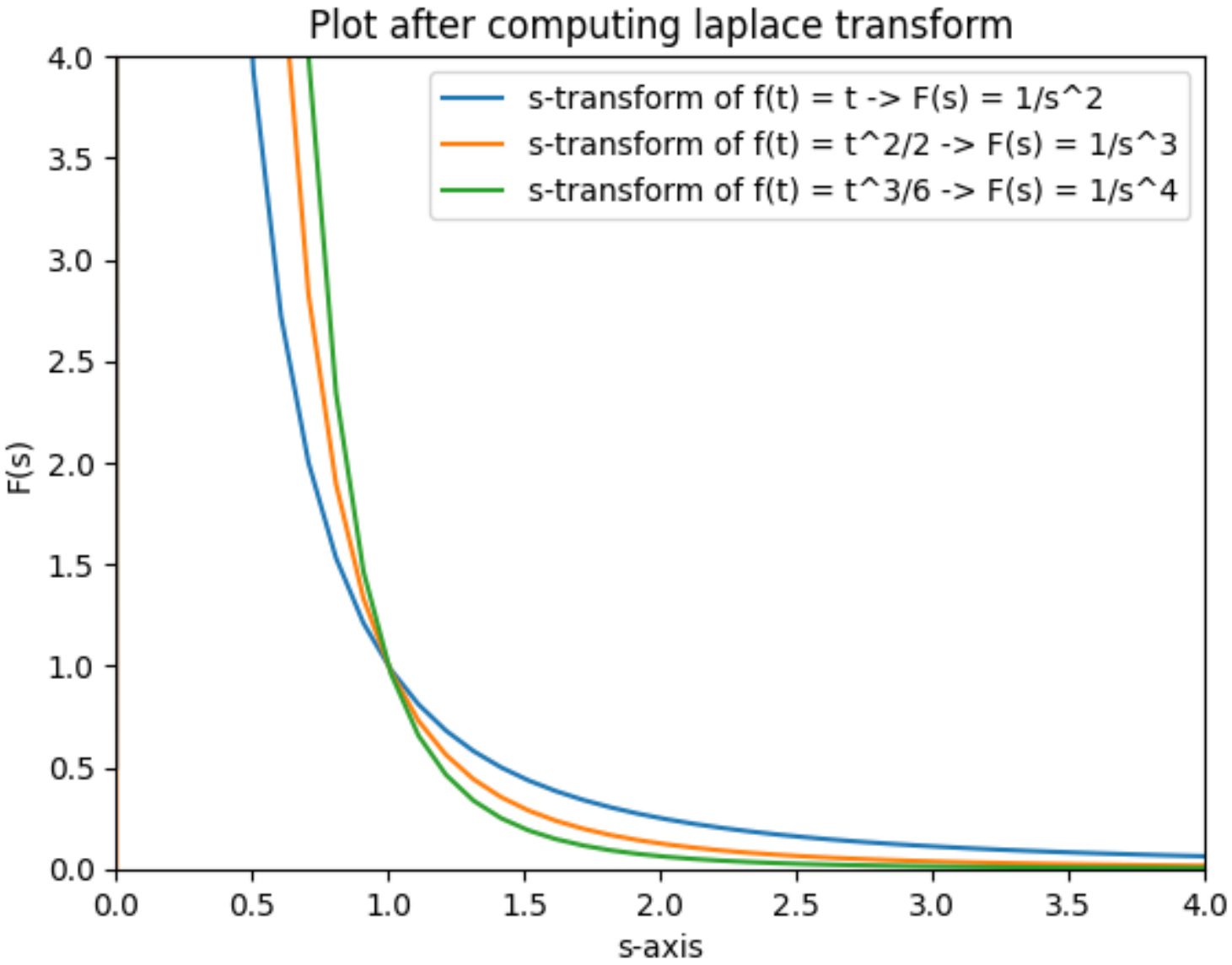
Python Code - Computing and plotting laplace transform of given signals I

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import integrate
import math
def integrand(t, s, a):
  return (t**a)*np.pow(np.e, -s*t)/math.factorial(a)
n = 100
S = np.linspace(0, 10, num = n)
lt_1 = np.zeros(n)
lt_2 = np.zeros(n)
lt_3 = np.zeros(n)
for i in range(0, n):
  lt_1[i],_ = integrate.quad(integrand, 0, np.inf, args=(S[i], 1)
      )
  lt_2[i],_ = integrate.quad(integrand, 0, np.inf, args=(S[i], 2)
      )
```

Python Code - Computing and plotting laplace transform of given signals II

```
  lt_3[i],_ = integrate.quad(integrand, 0, np.inf, args=(S[i], 3)
      )
plt.plot(S, lt_1, label = 's-transform of f(t) = t -> F(s) = 1/s
    ^2')
plt.plot(S, lt_2, label = 's-transform of f(t) = t^2/2 -> F(s) =
    1/s^3')
plt.plot(S, lt_3, label = 's-transform of f(t) = t^3/6 -> F(s) =
    1/s^4')
plt.xlim(0, 4)
plt.ylim(0, 4)
plt.legend()
plt.xlabel('s-axis')
plt.ylabel('F(s)')
plt.title('Plot after computing laplace transform')
```

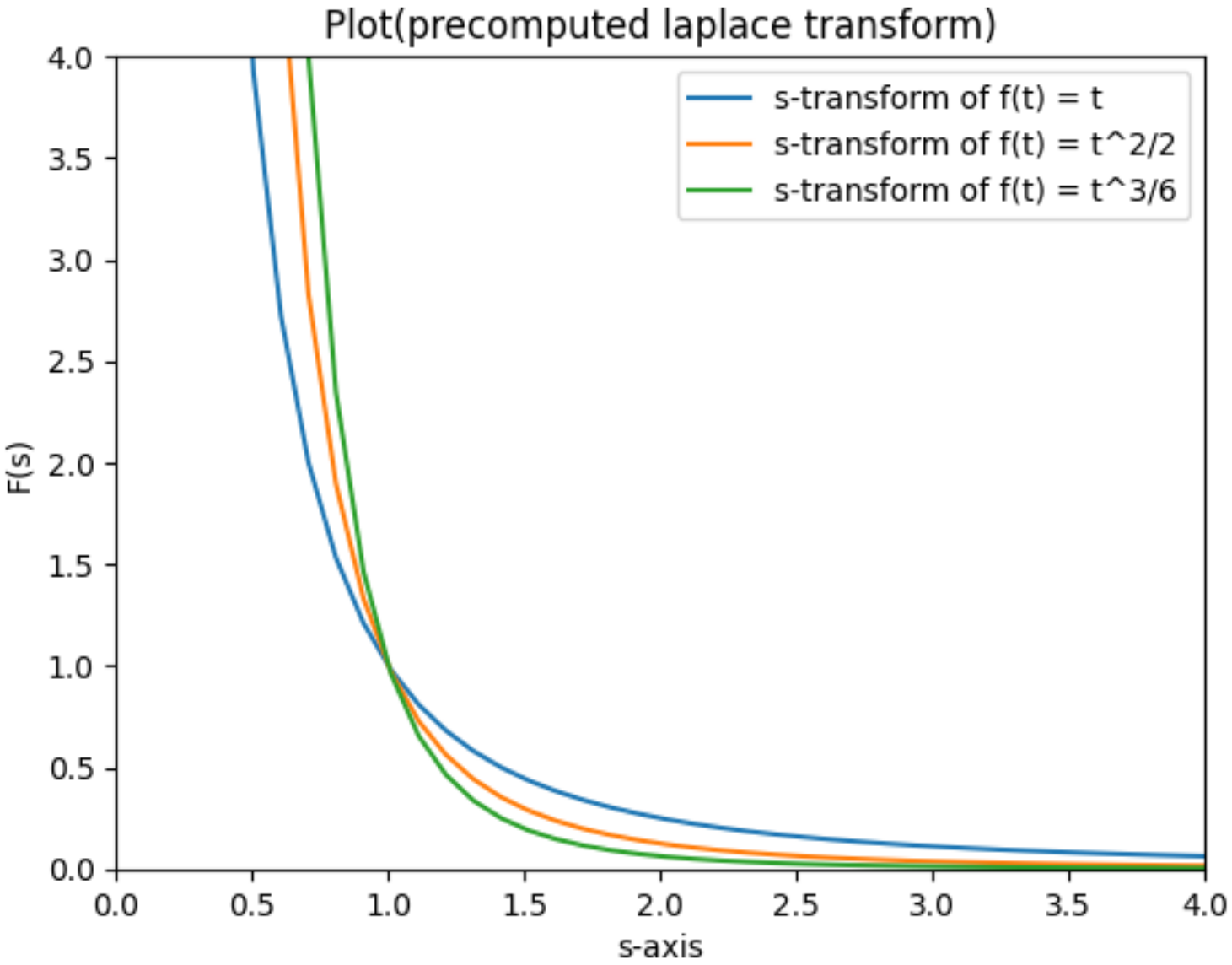
Plot for computed s-domain signals: $\frac{1}{s^2}, \frac{1}{s^3}, \frac{1}{s^4}$



Python Code - Plotting $1/s$, $1/s^2$, $1/s^3$ (without perfoming laplace transform) I

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import integrate

n = 100
S = np.linspace(0, 10, num = n)
lt_precalculated1 = (1/S)**2
lt_precalculated2 = (1/S)**3
lt_precalculated3 = (1/S)**4

plt.plot(S, lt_precalculated1, label = 's-transform of f(t) = t')
plt.plot(S, lt_precalculated2, label = 's-transform of f(t) = t
    ^2/2')
plt.plot(S, lt_precalculated3, label = 's-transform of f(t) = t
    ^3/6')
plt.xlim(0, 4)
```

Python Code - Plotting $1/s$, $1/s^2$, $1/s^3$ (without perfoming laplace transform) II

```
plt.ylim(0, 4)
plt.legend()
plt.xlabel('s-axis')
plt.ylabel('F(s)')
plt.title('Plot(precomputed laplace transform)')
```

Plot for: $\frac{1}{s^2}, \frac{1}{s^3}, \frac{1}{s^4}$ (direct plot)